

1 Solution — GIAM 6.2

Problem 4

1.1 Problem

For each of the five properties, name a relation that has it and a relation that doesn't.

1.2 Setup

Throughout, X is a non-empty set and $R \subseteq X \times X$ is a relation. The five properties:

- **Reflexive.** For every $x \in X$, $(x, x) \in R$.
- **Irreflexive.** For every $x \in X$, $(x, x) \notin R$.
- **Symmetric.** $(x, y) \in R \implies (y, x) \in R$.
- **Antisymmetric.** $[(x, y) \in R \text{ and } (y, x) \in R] \implies x = y$.
- **Transitive.** $[(x, y) \in R \text{ and } (y, z) \in R] \implies (x, z) \in R$.

The examples below all live on familiar sets — $\mathbb{R}$ for the order relations, an arbitrary set for $=$ and $\neq$.

1.3 Reflexive

- **Has it.** $\leq$ on $\mathbb{R}$: $x \leq x$ for every real x .
- **Doesn't.** $<$ on $\mathbb{R}$: $x < x$ is never true, so the diagonal is missing entirely.

Note that “doesn't have reflexivity” here is in fact the stronger property *irreflexive*. The two are not negations of one another: a relation can fail to be reflexive without being irreflexive — for instance, $\{(1, 1)\}$ on $\{1, 2\}$ contains the diagonal point at 1 but not at 2.

1.4 Irreflexive

- **Has it.** $<$ on $\mathbb{R}$: $x < x$ never holds.
- **Doesn't.** $\leq$ on $\mathbb{R}$: every diagonal pair (x, x) lies in the relation.

1.5 Symmetric

- **Has it.** $=$ on any set: $x = y \implies y = x$. (Or " x has the same astrological sign as y " from Problem 2.)
- **Doesn't.** $<$ on $\mathbb{R}$: $1 < 2$ but $\neg(2 < 1)$.

1.6 Antisymmetric

- **Has it.** $\leq$ on $\mathbb{R}$: if $a \leq b$ and $b \leq a$, then $a = b$.
- **Doesn't.** $\neq$ on any set with at least two elements: pick distinct a, b . Then $a \neq b$ and $b \neq a$, yet $a \neq b$ — the antisymmetric implication fails on this pair.

1.7 Transitive

- **Has it.** $<$ on $\mathbb{R}$: $a < b$ and $b < c$ give $a < c$.
- **Doesn't.** "Is a parent of." If Alice is a parent of Bob and Bob is a parent of Carol, then Alice is Carol's *grandparent*, not parent. (A formal version: the relation $R = \{(1, 2), (2, 3)\}$ on $\{1, 2, 3\}$ contains $(1, 2)$ and $(2, 3)$ but not $(1, 3)$.)

1.8 Remark — A Compact Cheat Sheet

A handful of canonical relations cover all five properties in both directions. Take any non-empty set X ; for $\leq$ and $<$, take $X = \mathbb{R}$.

Relation	Reflexive	Irreflexive	Symmetric	Antisymmetric	Transitive
$=$	Yes	No	Yes	Yes	Yes
$\neq$	No	Yes	Yes	No	No
$\leq$	Yes	No	No	Yes	Yes
$<$	No	Yes	No	Yes	Yes

Table 1.1

A few patterns worth noting:

- **Strict vs. weak.** The strict relations ($<$, $\neq$) are irreflexive; the weak relations ($\leq$, $=$) are reflexive. Reflexive and irreflexive form a clean dual axis here, even though in general they are not negations.
- **Equality is the unique all-yes row.** This is no accident: a relation that is symmetric *and* antisymmetric must be a sub-relation of the diagonal, so equality (and its sub-relations) are the only candidates for the all-yes row (cf. Problem 2’s remark).
- **$<$ is antisymmetric vacuously.** No pair (a, b) has both $a < b$ and $b < a$, so the implication’s hypothesis is unsatisfiable and the column entry is “Yes” by default — the same trick that made “smarter than” antisymmetric in Problem 1.
- **$\neq$ is the most lopsided.** It fails three of the five (reflexive, antisymmetric, transitive) yet has the other two (irreflexive, symmetric) — handy when one needs a relation that emphatically isn’t an order or an equivalence.